



DO-IT-YOURSELF: CIRCUIT IDEA

and LED18 flashes slowly.

In this circuit, pin 5—a voltage control pin—is used. This pin is rarely used, and when not used, it is generally grounded through a $0.01\mu\text{F}$ capacitor.

Here, pin 3 of IC18 is connected to pin 5 of IC19. When the circuit is powered, both LED17 and LED18 flash fast, almost synchronously. This is because the output square wave of IC18 astable multivibrator with higher frequency overrides and modulates the square wave of IC19 astable multivibrator with lower frequency through pin 5.

Now, if the situation is reversed and output from pin 3 of IC19 is connected to pin 5 of IC18, the phenomenon gets reversed. The flashing of both LED17 and LED18 starts. The flashing of LED17 being faster than that of LED18, the latter appears to be steady but soon both the LEDs turn off synchronously for some time and the cycle continues.

Experiment 10

In tenth experiment, as shown in Fig. 10, IC21 is wired as a simple oscillator. The internal voltage divider of IC20, consisting of three 5k resistors in series that internally create reference voltages for the internal comparators, is used here in part as a resistive path to charge and discharge capacitor C11.

Pin 3 of IC21 is connected to pin 5 of IC20. Pins 2 and 6 of IC21 that are tied together are connected to pin 1 of IC20. When the circuit is powered, LED 19 flashes. If pins 2 and 6 of IC 21 are connected to pin 8 of IC20 instead of pin 1 of IC20, the flashing rate of LED19 increases, as the resistance decreases.

Experiment 11

In the circuit for this experiment, shown in Fig. 11, IC22 is wired as a monostable multivibrator and IC23 as an astable multivibrator. Pin 8 of IC23, instead of connecting to V_{cc} , is connected to pin 3 of IC22.

When switch S14 is momentarily pushed, the output of pin 3 of IC22 goes high and powers IC23. Both the capacitors C12 and C13, which are identical, start charging almost simultaneously.

However, charging of C13 is comparatively slower as it charges through two resistors in series. The LED20 turns on but before C13 gets a chance to charge to more than $2/3$ of V_{cc} , C12 charges to more than $2/3$ of V_{cc} and IC22, which is wired as a monostable multivibrator, completes

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